

LINKING ACADEMIA WITH INDUSTRY: PARTNERSHIP APPROACH IN DIFFERENT UNIVERSITY SETTINGS IN KENYA

SUMMARY

The role of universities as key knowledge and innovation producers is widely acknowledged. Universities generate new knowledge, innovations and technologies that are transferable to industry and society for use through publications, consultations and knowledge exchange. Most of the universities have now embraced the third mission framework of economic research through research commercialisation, entrepreneurship, patenting activities as well as enlarging their business network. The public universities rely more on publicly funding. The active participation of public universities in patenting and searching for alternative funding sources for the internally generated knowledge and innovations has not been a major priority, although lately this is slowly changing. The private universities have had an aggressive thrust in commercializing their knowledge and innovations. The transfer of such knowledge and innovations have become a major income earning activity in addition to income from the student fees for private universities. There are mechanisms in existence for sharing and transferring these hard and soft skills with the private sector although there are claims that this has not yielded adequate economic returns to the society, thus the need to understand the mechanisms for enhancing university industry (U-I) linkages for both public and private universities.

The case study looked at the practice of industry-university partnerships in public and private settings. Jomo Kenyatta University of Agriculture and Technology (JKUAT) was selected for a publicly governed university while Strathmore University (SU) represented a privately governed institution. The University – Industry linkage remains a recognised challenge of leveraging university knowledge, innovations and expertise into private sector and vice-versa. The case study investigated the nature of the mechanisms used to build links with the private sector in each university giving particular attention to funding arrangements, governance structures and other institutional arrangements that have enabled or impeded these linkages. It examined the processes that continue to shape the development of the partnerships. More specifically, we looked into how the U-I linkage mechanisms are effective (including barriers and enablers) in supporting innovation, inclusive knowledge systems and achievements in promoting the Sustainable Development Goals (SDG).

Jomo Kenyatta University of Agriculture and Technology has focused on its niche in agriculture, and has managed to connect with partners advancing societal needs in this sector. While its public governance structure has played a role in terms of goals pursued, it has influenced to some extent the types of actors to link with and this sort of narrowed the university's ability to enhance research growth trajectory in university-industry linkages. JKUAT has focused more on creation of new knowledge e.g. through the establishment of a centre to promote ethno-botanical knowledge, yet the linkage to industry consumer remains weak and undefined in its strategic plans.

On the other hand, SU offers the opportunity to demonstrate a relatively recent development in the knowledge system whereby private universities are able to forge new patterns of partnership with private industry. The University has focused on its core mandate to advance business partnerships and has managed to roll out activities/services that focuses on

economic growth, industry innovation and infrastructure, affordable and clean energy through the various research centres. Strathmore University is striving to make the university a business hub as well as an education and research hub. They are increasingly trying to embed industry in the University to strengthen education industry synergy and research industry synergy, e.g. through establishment of a law firm within the law school.

The study revealed that, on a general front, these universities have succeeded in forging linkages between producers and users of knowledge. However more specifically, the effectiveness of such linkages are shaped significantly by certain niches and governance frameworks that either enable or impede engagements and mutual and inclusive actions and benefits. We have provided a summary analysis of the findings in Table 1 where we present the knowledge systems innovation and other findings. We present an opportunity to consider strengthening different pathways for research in the case for JKUAT while Strathmore University should consider the use of privately or publicly funded innovation funds to direct and enhance success with transfer of knowledge and innovations to business for sustainable social and economic development

Table 1: Case Study Summary Table

Case study description	
Knowledge System domain of interest	Connecting university research to the private sector and other users
Knowledge system innovations	Creating links between producers and users of knowledge through partnership
Knowledge system actors and resources	
Key knowledge actors	JKUAT: University and county governments, SME and communities focused on agriculture. Strathmore: University and medium to large scale industry across multiple sectors
Key knowledge sources	Formal knowledge generated by university research and enterprise related knowledge from industry and in JKUAT indigenous knowledge.
Financial resources and investment	JKUAT, public investment in education and research and small scale investment in specialised training and technology transfer centres. Strathmore, revenue from student enrolment invested in specialised research centres to service and partner with industry.
Knowledge system functionality¹	
Nature of networks	JKUAT: mainly with research organisations, county governments and communities Strathmore: medium and large scale industries and the international research donor community
Nature of knowledge flows	JKUAT: uni-directional, top down, research driven Strathmore: interactive around co-creation and co-innovation

¹ This implies the relationships and networks (formal and informal) through which knowledge is transmitted and combined and used for innovation, the institutional arrangements, governance arrangements, policy settings that enable and direct the use of knowledge for innovation.

Enabling condition (organisation)	JKUAT: Conforms to a classic research and extension organisational setting that encourages a focus on academic excellence and transfer of knowledge through other public agencies. Strathmore: Privately owned with private sector governance, actively focused on business service, driven by alumni networks in the business community. Culture of starting lean to understand new domains and spinning of research centres to facilitate partnership with the private sector. Organisational innovation such as the deployment of knowledge brokers to identify partners has been part of the success.
Enabling conditions (wider)	A range of systemic challenges that weaken innovation by industry, including policy and regulatory issues, as well as under developed innovation capacity in SMEs that dominates in many industries particularly agriculture and makes partnership challenging.
Knowledge systems performance.	
Actors success narrative	JKUAT: Academic excellence (education and research). Highly successful in using tissue culture to distribute disease free planting material to local communities. Trained farmers and local community in use of dairy processing technology developed by the university. Strathmore: Explicitly target industry as a consumer of their products and also as a basis for setting research activities. Industries are continuously approaching SU to develop solutions together, particularly energy research and the internet of things. Research has to be for impact (of all sorts) in line with the University's vision of setting the next technology frontier of industry, particularly ICT.
Actors failure narrative	JKUAT: Doesn't target industry as a primary consumer but has embraced an open knowledge use Strathmore: Aligning everyone in the U-I partnership to see the bigger picture due to varying interests.
Directionality and explicit mechanism to focus on SDGs	JKUAT: set up with a specific focus on small holder agriculture and supporting local communities, partnering with devolved country governments on local development priorities. Recently establishing, with support from JICA, Sino-Africa Centre for promoting Indigenous, ethno-botanical knowledge. Strathmore: Explicitly try to blend business and community interests. Established Kenya Climate innovation Centre with strong focus on supporting SME's launching climate smart innovation. Other relevant specialised Centres focus on issues related to sustainability, e.g. iLab and energy research centre,
Knowledge system political economy distortions, lock-ins and contextual factors	
Power dynamics and governance	JKUAT: Strong government control, limited autonomy in decision making, political influence on appointments and management. Strathmore: Flexible governance guided by the needs of industry and very strong private sector/ market driven governance and agenda setting
Lock ins	JKUAT: Historical framing as a technology development and extension agency, patterns of capability related to this, and a partner network reflecting historical perspective. Strathmore: Few lock-in, but has a very strong market facing trajectory because of history and current patterns of governance and stakeholder engagement. The agenda is not blind to the social and sustainable agenda, but is always filtered through a market lens.
Reflections and learning	
What is working well	The two case studies differ in governance and operations targeting different sectors of the economy. The knowledge generation is linked to specific consumers. Strathmore has very strong culture and vision of working with the private sector and this enabled them to establish research centres specifically as collaboration nodes with industry. It has also developed non-traditional capability such as

	brokering of knowledge and innovation. Energy and climate innovation centres are particularly powerful in stimulating sustainability ,enabling entrepreneurship, and in driving economic success more generally. JKUAT, deals more with agriculture with less focus on formal industry links, focusing instead on communities and the SME sector. Given the nature of agriculture in a developing country, needs to follow a variety of innovation pathways which can be translated to private sector led commercialisation. Collaboration with devolved county governments is particularly successful as well as collaboration with some training centres.
What needs more attention	Strathmore, while performing well in a general sense, targeted funding by the public sector on contractual basis would give greater incentives to find a better balance between the economic, social and sustainability benefits it impacts to society. JKUAT is yet to arrive at a model research – user interface needed to deploy the range of research products and expertise that it holds. Capacity building, institutional development and creating space for new models and partners of partnership could help an organisation that is already well directed towards the sustainability agenda
Knowledge system innovation [fringe approaches and novel process]	Strathmore: Trying to make the university a business hub as well as an education and research hub. The synergy created increasingly embraces industry in the University to strengthen education-industry and research-industry link JKUAT: Establishment of a centre to promote ethno-botanical knowledge has potential to renovate agriculture if commercialized.
Opportunities and options (Practical steps/approaches?) THE HOW?	JKUAT. Search desperately for solutions that transfer knowledge and innovation into use conundrum. Potential for investment in testing, adaptation and learning about different pathways to research. Strathmore: Bid for publicly funded projects to direct successful research with business engagement and innovation towards social and sustainability issues.

1.0 INTRODUCTION

In a complex world where knowledge produced by University needs to be utilized, there has been pursuit of linkages with the end users including industry. Knowledge production is increasingly being understood not just as contribution to the research and education systems but actually how the knowledge generated by higher institutions of learning is put into use. To this end higher institutions partnerships linking to the industry is gaining traction. The industry is increasingly recognising the role these institutions could potentially play in contributing to research use and innovations in general. This growing appetite for university industry linkages has been witnessed by a number of both public and private institutions in Kenya.

Bolo et al. (2015) among other scholars introduces this linkage via a concept around third mission framework. The third mission framework demonstrates the Universities orientation towards “Innovation”, “entrepreneurship”, “extension” or “community outreach”. One way to ensure that the institutions engage in this mission is through stakeholders’ engagement through partnerships. Our case study explored how these “third mission” mechanisms work in practice by looking at industry-research linkages and specifically the role of partnerships in facilitating effective linkages between university and industry. We explored these partnership types in a model public and private universities. We have generally looked at how partnerships are formed and linkages with the industry/universities formed and strengthened. What types of partnerships are pursued? What are the knowledge sources and diversity in them? What are the implications they have on sustainable development outcomes? How is success measured versus failure narrative? This article responds to these questions in public and private university settings outlining how these partnerships are framed and proposing options around how we can learn from such interventions and investments to support both public and private knowledge producing institutions which are different in their configuration, orientation and design.

Both the public and the private Universities have managed, although in different degrees and in different ways, to link their research to research users. These institutions have capitalised on their various strengths including focusing on their niches to advance various partnerships.

Strathmore University for example has a very strong culture and vision of working with the private sector due to its origins, founded under a premise to pursue business courses and has grown its alumni network across this space hence using this to forge strong networks. This is echoed by Este and Patel (2007) who postulate that commercial orientation of university research affects U-I linkages as well as its founding objectives and technology transfer experience. It is these orientation that then affects behaviour of the researchers/lecturers as it also shapes the institutions.

Both universities are therefore anchored around the specific niche they were established to pursue. Their culture of starting lean and expand from faculties to research centres as collaboration nodes has enhanced industry-university linkages. Investments on non-traditional capability such as brokering has also enabled linkages. These centres have become powerful in stimulating sustainability to support entrepreneurship, and other in driving economic success more generally.

On the other hand JKUAT's strength lies not on the formal industry partners but more on agriculture and agri-based chains focusing on communities, extension and the Small to Medium Enterprise (SME) sector and the County governments through its training centres.

Generally, all these institutions are advancing U-I linkages based on their niches. Competition between both public and private universities for public as well as private contracts has increased, resulting in an inevitable change in knowledge production with third mission activities oriented towards a direct socio-economic impact. However, university–industry interaction does not involve only transferring knowledge from the former to the latter; but also helps academics to access industry skills and facilities, develop interesting research questions, and provide improved understanding of research applications in industry. Building on Bolo et al. (2015), we reinforce the recommendation around the need to position Universities' centres of excellence in areas of competitive strengths and ensure that this is widely shared with the business community.

2.0 HISTORY, CONTEXT AND OPERATIONAL DETAILS

2.1 Jomo Kenyatta University of Agriculture and Technology (JKUAT)

Jomo Kenyatta University of Agriculture and Technology (JKUAT) was started in 1981 as a Middle level College (Jomo Kenyatta College of Agriculture and Technology (JKCAT) by the government of Kenya. The college was then declared a constituent College of Kenyatta University through a legal notice, Kenyatta University Act (CAP 2010c). The name was then changed to Jomo Kenyatta University College of Agriculture and Technology before being fully established as a University through JKUAT Act, 1994 and inaugurated in 1994. The University was purposed to, among other functions: participate in the discovery, transmission and preservation and enhancement of knowledge and to stimulate the intellectual participation of students and researchers in the economic, technological, agricultural, professional and cultural development of Kenya. The University has and continues to play an effective role in the development of agriculture and technology in conjunction with the industry through research, production and innovations, provision of extension and transfer services as well as establishing linkages.

The university operates under five (5) thematic research centres namely; the Institute of Energy and Environmental Technology (IEET), Institute for Biotechnology Research (IBR), Sino-Africa Joint Research Centre (SAJOREC), Water Research and Resource Centre (WARREC) and Sustainable Materials Research and Technology Centre (SMARTEC). These thematic research centres have income generating units as technology transfer centres attached to specific departments interfacing with the industry and the public through training and selling of products generated. These units include the chemistry products centres, food technology centre, JKUAT nursery school and the university bookshop. Being a public University, it is publicly funded through government capitation, as well as research funds from National Research Fund (NRF) (CPS, 2018) and donor sources. Additionally there are other small scale investments in specialised trainings from the five technology transfer centres as supplementary sources of income.

The University has positioned itself as a knowledge producer with a strong focus on agri-food-based value chain type of linkages. Majority of the partnerships pursued are in agriculture. These partnerships are established through call for proposals to link University-private sector and research. One example is the Sorghum Value Chain Development Consortium that targeted development of sweet sorghum value chain with an investment of about USD 2M. Secondly, partnerships are built through targeted funding in consideration of impacts on social-economic development as well as emerging opportunity through donors and well-wishers who wish to collaborate.

The partnerships focus on thematic areas such as: Research collaboration; production partnerships (commercialization of innovations); extension services in collaboration with media houses through TV programmes; and national priorities such as agriculture and food security, health, engineering, and information technology. These partnerships have mostly been successful in areas like the tissue culture banana that involved innovative extension models with partners who supported the value chain approach by training farmers.

2.2 Strathmore University (SU)

Strathmore University, a private university, was first granted authority by the government of Kenya in 2002 to offer flagship undergraduate programs in Commerce and Information Technology and later in 2008, awarded a Charter by the Government of Kenya with full legal recognition to operate as a University. As a private university, the institution funds its own programmes through student enrolment. The University also receives funding from other non-state research funding sources and is the leading privately funded university (CPS, 2018). In addition, revenue is generated from short trainings, professional courses, specialised courses and industrial training. It has grown over time to offer academic programs like masters (started in 2012), and recently in 2017 has started offering PhDs.

Strathmore University currently hosts 16 research centres actively dedicated for research. These centres are spin-off from faculties that have previously been centred on knowledge production. These centers include; (i) Kenya Climate Innovation Centre (ii) Strathmore Energy Research Centre (iii) Strathmore Extractives Industry Centre (iv) Strathmore Enterprise Development Centre (v) @iLabAfrica Research & Innovation Centre (vi) Centre for Research in Therapeutic Sciences (vii) Center for Intellectual Property and Information Technology (viii) Strathmore University Press (ix) Strathmore Centre for Value Investing (x) Strathmore Risk Management Centre (xi) Strathmore Center for Sustainable Leadership (xii) Center for Research in Organization Work and Family (xiii) Strathmore Institute for Healthcare Management (xiv) Strathmore Institute for Public Policy & Governance (xv) Strathmore African Centre for Business and Human Rights (Xvi) Strathmore Law Firm.

These research centres together with various faculties are involved in developing partnerships both locally and internationally to address the vision of the University. Partnerships are pursued through the various research centres together with the faculty members. The focus areas of these partnerships are pursued along the research centres' mandates that range from joint ventures for research implementation and dissemination to research governance through agenda setting in various sectors. For instance, the iLabAfrica research centre is currently partnering with Safaricom Limited (a mobile telecommunication

company) to train the company's staff on emerging technologies, an area with limited expertise in the country. The University has also focused on value-addition for the society by advancing knowledge locally through a partnership with Kenya Revenue Authority (KRA). In the year 2017, 22 students were deployed to various stations across Nairobi to assist with tax file returns. Learning was two way and mutually benefiting to the society. While the masses/communities benefited by acquiring hands on knowledge, the activation exercise enhanced the students' knowledge on taxation, blending classroom knowledge with the job market expectations.

This demonstrates interactive flows of knowledge in theory and in practice relevant for communities. This approach reinforces Bolo et al. (2015) sentiments around how Universities churn out new research outputs which could be in the form of technologies, processes, services, that could benefit the private sector and communities, and whereas on the other hand the private/communities sector seeks solutions to their routine challenges as well as trying to break into new areas. These sorts of partnerships are therefore breaking into these confinement and demonstrating how such partnerships are important in co-creation and co-development of innovations.

It can then be deduced from the analysis above that both SU and JKUAT have similarities in their approach to partnerships. Partnerships are pursued through ongoing series of collaborations between two individual partnering institutions or through two concurrent large scale projects involving industry and academia at the same time. Others are through competitive project based funding, formalized through formal institutionalized partnership while others are informal partnerships that are eventually formalized. These partnerships are usually with local and international partners but based on geopolitical contexts with related interests/mutual benefits involving individual network carried into the partnerships building. As observed, the overreliance on project based competitive funding has been very detrimental to incentivising third mission activities.

As a result, SU usually pursues knowledge generated while maintaining a network of small and medium scale industry, government and the international community. On the other hand, JKUAT maintains a network of research organisations, county governments, centres and communities. What has been observed from the respondents is also the export of knowledge especially from SU through positions as professorial adjunct positions by the partners although emphasis is always on local rigor. SU has managed to therefore stay locally relevant while advancing their knowledge and expertise internationally.

3.0 GOVERNANCE, ACTORS, OBJECTIVES

3.1 Public University- JKUAT

The university is governed through a government appointed Council of not more than nine persons responsible for policy decisions and implementation. The Board of Management composed of principals of colleges, academic directors make decisions on day-to-day management and operation of the institution. Some of their roles as outlined by respondents include ensuring graduates are churned out well, ensuring patents are filed and copyrights

registered with the Intellectual Property Rights organization as well as conducting products development, although only at the prototype level.

Research aspects of the University are domiciled in the Research, Production and Extension (RPE) Division, one of the four Divisions in JKUAT. The Division was set up when JKUAT, then JKUCAT, became a constituent College of Kenyatta University in 1988 under the headship of a Deputy Principal. In 1994 when JKUAT attained the charter to operate as a fully-fledged University, the headship of the Division changed to Deputy Vice Chancellor. The Deputy Vice Chancellor, in carrying out the various mandates of the Division, is supported by a Registrar and various Directorates, Departments, Research Institutes and Centres. The office of the Deputy Vice Chancellor (RPE) coordinates such services as: research including fundraising and dissemination; production and income generating activities; extension and technology transfer services including shows and exhibitions, and linkages services including community collaboration. Although very centralised and a good practice in coordinating knowledge transfer, this is not what is in practice but rather smaller departmental offices that coordinate partnerships are very bureaucratic in nature thus limiting such partnerships.

The University leadership is domiciled by mostly academic staff who have prioritised academic excellence and transfer of knowledge through other public agencies. This linear approach has encouraged uni-directional top down driven research. A shift in governance arrangement for universities like some of the private universities where the private sector/business people are allowed in leadership positions may allow interactive partnerships where the private sector influences some of the university decisions. As proposed by the respondents, a clear policy direction is needed around how managers are chosen in public universities so as to incorporate private sector players in leadership or academic individuals with proven track record in business.

Jomo Kenyatta University of Agriculture and Technology collaborates with diverse actors and partners in research and innovation. These include county governments, private sector players, Civil Society Organizations (CSOs), research institutions, and community groups. The actors engage in activities around the provision of business models, financial resources, management skills and capacity building as well as value chain identification and support mostly in the agricultural sector.

The areas of partnerships for JKUAT include: collaboration in offering academic programmes, collaborations for research development and technology transfer and collaboration with donor bodies. Collaboration activities cover such areas as: commercialization, outreach, project management, dairy farming, staff and student exchange, capacity building, student attachment, research funding, patent and co-authorship, curriculum development, training programmes, joint research, extension, student career promotion, conferences, consultancies, artisan training, licensing, financing of collaborative programmes, value addition, entrepreneurship and enterprise development, export enhancement, shows and seminars, education, exhibitions and agricultural demonstrations, business planning and feasibility studies, development of built environment, young graduate trainee programmes, participation in career fairs, information dissemination, establishment of science park, solid waste management systems, market driven innovations, assisting and supporting pilot projects, and sharing of physical facilities.

These partnerships have been enabled through relationships built over time like the case of JICA. Being a public funded institution some partnerships are aided by the Government initiatives especially where public universities are requested to spearhead linkages with industry/apply for hosting agreements etc. Therefore JKUAT conforms to a classic research and extension organisational setting that encourages a focus on academic excellence and transfer of knowledge through other public agencies. In other words, the target is not necessarily commercialisation via private sector consumption but wider public consumption.

Partnerships in JKUAT has yielded a number of benefits both in the remit of intellectual enhancement as well as infrastructural development. For instance, the JICA partnership has enabled the University to establish learning infrastructure including buildings and machinery but most importantly, capacity building. Other benefits have been experienced by individuals who have forged those partnerships as they receive recognition during graduation speeches and use these recognition for certificates of merit. However, JKUAT interviewees indicated that the institution has not yet worked out how prime movers are rewarded. The challenge of finding the right incentives for third mission activities has hampered linkages calling for greater institutional and political leadership.

While most respondents appreciate these positive outcomes, there were nonetheless concerns that some partnerships (an example was given in the case of JICA) mainly favour deployment of foreign knowledge at the expense of local relevant technological knowledge.

3.2 Private University - Strathmore University

Strathmore University is a fully fledged university with a Vice Chancellor, three Deputy Vice Chancellors (DVC) academic and student affairs, DVC planning and development and DVC research), the University Secretary, a number of senior management staff, the University Council, the Management Board, the Academic Council, and various committees. The university structure is in such a way that majority of their board members are representatives from the industry hence their agenda setting, and orientation lean more towards training students for the industry. These members have influenced heavy investments from non-state research funding.

The key knowledge actors in formation of partnerships with industry in the case of SU include other universities and medium to large scale industry across sectors. These partnerships are advanced across the different faculties with the different research centres providing evidence-based knowledge that drives new ideas and technologies essential in propelling the future of society and humanity while varying in terms of the thematic areas, and duration as well as disciplinary underpinnings. The partnerships work along modalities where formal² knowledge is generated by the University and enterprise³ related knowledge is generated by the industry. The interaction is not only premised on universities applying their research but

² Knowledge produced by research institutions, usually inscribed in printed and digital artefacts. This knowledge in this context often has some externally imposed criteria and agendas.

³ This is knowledge that workers/employees and the industry possess in a particular enterprise, it is very diverse and exists and is more or less generated from the workers from their experience.

also for additional income, as the main funding is from students' fees. Also the interaction results in access to industry skills and facilities and keeping abreast of industry needs.

This university interaction with industry has exposed them to a wide range of technological problems identified by industry, opening an array of research avenues that would not have emerged had researchers remained within the boundaries of university research. In other words, the research agenda is generated either from the university and/or from the industry. In some cases the industry injects catalytic initial funding to try the concepts or translate concepts to implementable projects. This has enhanced a network where collaborations and partnership types are mutual. In other words, the case of Strathmore indicates that effective U-I linkage and associated knowledge flows are enabled through co-creation and co-innovation.

Establishing platforms to enhance knowledge flows beyond the boundaries of the University is also a key enabler. For instance, the SU has deliberately established some outreach centres to enhance access and up-take by both SME and large ones. These include the [Utawala Applied Research Institute](#), a County governance resource centre that provides relevant information and data for designing and delivering County development priorities and projects. Information accessible in the centre include latest research and innovations, business ideas, expert analysis and support to expert advice, training and capacity building and projects feasibility assessment.

The 16 research centres (described on page 8 above) spanning across different fields are involved in at least the establishment of partnerships to engage students and faculty in research which has critical economic, societal and environmental impact nationally, regionally and internationally. The growth of the schools has expanded the scope of teaching faculty and research centres. These centres have emerged progressively based on the business orientation of university that has greatly affected linkages and the technology transfer experience of the early established research centres. The centres have employed the remit of a more market oriented direction for example the internet of things centre. The practices established by these research centres strongly influence the disposition of lecturers and researchers to set up networks with users of their research.

All the centres are key in driving sustainable development outcomes with two key centres that stand out. These include Strathmore's iLabAfrica, ICT and Innovation centre, a Centre of Excellence in ICT Innovation and Development established to address the Millennium Development Goals (MDGs) in 2011 and contribute towards Kenya's Vision 2030. The centre strives to promote technological innovation and provide business support structures and policy direction to harness the potential of ICT as a tool for sustainable development. Since its establishment, the centre has attracted enormous interest from other institutions building a partnership with over 20 institutions (in line with SDG No 17): both academic and industrial players locally, regionally and globally in the areas of internet of things and building new technology solutions. The partnerships are beneficial in that SU has received industrial support that includes curriculum development for a program that also offers scholarships, setting up of the business incubators, mobile boot camps as well as jointly conduct research and develop policies on Kenya's innovation and further develop capacity on spectrum management skills.

Other areas include development of renewable energy sources by rural communities in a sustainable way. The Energy Research Centre (SERC) is one example that carries out high-quality research and consultancy services. The centre provides professional training, laboratory testing and project development for the renewable energy sector in Kenya serving the Government, private sector and the general public.

Contrary to the JKUAT case where partnership has enabled development of intellectual capability and research infrastructure, the benefits under Strathmore are more oriented towards establishing entrepreneurship through greater access to data by investors and academics e.g. the Pangea accelerator hosted at the University. The Arnjawalla and Khan partners who are hosted at Strathmore law school have built a library infrastructure for the school. The staff at the law firm are able to access the books while the university stocks up the library. Most importantly, the framing of U-I partnership has led to creation of significant employment opportunities with *“60-70% of hired students in Arnjawalla and Khan are from the Strathmore law school”*. Strathmore’s Annual 2017 report indicates that 80% undergraduate students and 77% of the diploma students were employed by the private sector.

Other developments include the Strathmore University’s establishment of County pro software for revenue collection to curb corruption at the county level. The SU students helping communities file tax return demonstrates the University and Industry’s shared commitment to responsive research that balances rigour and relevance.

Overall, it’s worth noting that the U-I linkage as pursued by SU appears not only to provide job linkages to students to apply their skills but also enhances the capability of students to engage with practical challenges through effective knowledge deployment. The latter provides a robust opportunity for SU to capture the market with its research and knowledge through capability building.

4.0 PERFORMANCE AND ACHIEVEMENTS OF THE PARTNERSHIPS

4.1 Performance of JKUAT partnerships

Most of the successful partnerships in JKUAT as mentioned by the key informants have emanated from initiatives from both individuals and institution wide. Also, the setup of the University is with a specific focus on small holder agriculture and supporting local communities, through partnerships with devolved country governments on local development priorities. This strategic niche area for JKUAT has been further emphasized by the establishment of Sino-Africa Centre for promoting indigenous, ethno-botanical knowledge.

There are certain clear examples where individual efforts have resulted in useful positive outcomes. An example is the partnership with the Chinese Academy of Sciences that resulted from a post doctoral student who attended the Sino Africa Joint Research Centre and won an award that has put up a 3 billion KES botanical garden in JKUAT that has a conference centre together with research labs to promote indigenous knowledge in Kenya. Further, the University has managed to sustain some long term partnerships with the JICA case running for over 41 years. This is a key achievement for JKUAT in terms of long term relationship

building. The University also prides itself around how it has managed to maintain academic excellence around education and research in agri-value chains with a key highlight on the tissue culture banana. This innovation has managed to create impact by the university distributing disease free planting material to local communities and this has been adopted widely. Through the technology transfer centres, the institution has managed to train local farmers in the use of processing technologies in the dairy technology, fruit processing technology and food production principles in general.

However, the University has not done well in the quest to commercialise research products. At the time of conducting this case study (2019), the University had over 50% of its research outputs not commercialised nor scaled up to production level mostly in the agricultural arena. Este and Patel (2007) argue that in most cases, academic R&D supported by industry is often geared towards getting up to date knowledge, obtaining access to students and faculty and finding solutions to specific problems and maybe could be the reason JKUAT has failed to move to commercialisation. The University has also failed to move into manufacturing sector a blame put on its governance mechanism and its linearity around research and lock-ins arising from historical framing as a technology development and extension agency leading to patterns of partnerships that reflects this history.

More broadly, the University framing of partnership has potential to promote various SDGs including food security, gender, and technologies. While these benefits are supportive to the public good, the process could be improved through strengthening linkages to the industry so as to enable more economic returns and poverty reduction. Doing so would require the University to make intentional efforts in mainstreaming the linkage in its structures (i.e. management system) as well as research choices.

4.2 Performance of Strathmore University Partnerships

As already highlighted, SU is keen to promote knowledge flows in the areas of energy and internet of things. A key feature of this is that SU has managed to create training programs and collaborate with industries while continuously approaching the industry to develop joint solutions. Such include the current partnership with Safaricom Limited (Telecommunication Company) where the University is championing training of the company's staff in the development of software applications.

Additionally, SU appears to be focused on catalysing positive outcomes as a key determinant of success. The partnerships that has contributed to a broader common goal of social justice and poverty alleviation were stated as some of the successful partnerships in linking with industry. The SU have demonstrated this through the development of County pro software that is delivering impact by revenue collection curbing cases of corruption at the county level. The university is also embracing a culture of vision setting to be the next technology frontier of industry, particularly ICT.

The University equates this success in their ability to sustain productive relationships beyond research duration. Even when the projects and the collaborations have technically ended, they have continuously kept relationships which has led to other engagements/referrals. They indicated that most of their partnerships have generated trust to the institutions and now the process seems very effortless. One of the partners quoted "*we approach Strathmore*

on collaboration based on the trust we have with them and they do deliver, they strive to have a brand internationally". Therefore building trust also emerges as a key enabler of sustainable partnerships that might catalyse U-I linkages.

Implications for sustainable development

Strathmore works on a premise that most of their training is skilled based and as knowledge producers they have special roles in advancing the SDGs. They have mastered the art of localizing research (by conducting research focused on solving problems and thus necessitates many partners to approach them), they also focus on regionalization of their research, and finally internalization through human formation, knowledge production and innovation hence the myriad of partners they have.

Through their research model, the University is explicitly trying to blend business and community interests. This has been demonstrated through the establishment of Kenya Climate innovation Centre with strong focus on supporting SME's launching climate smart innovation. Other relevant specialised Centres focus on issues related to sustainability including the establishment of sustainability leadership centre.

5.0 REFLECTIONS AND LEARNINGS

Linking industry with academia is understood differently by different actors. In summary and from the universities' perspectives, it is about universities listening to the industry for information to develop solutions but also the industry setting the agenda and research needs for the University to address. Generally, as observed by the respondents, failure by both public and private universities to pursue partnerships with industry is due to historical facts that the country's knowledge systems is developed around where innovations have previously emanated from, in this case, foreign firms.

The Kenyan-grown industry as currently constituted has a tiny industrial base save for devolution that has increased the number of actors in the knowledge system. Most of the industries in Kenya are subsidiaries of established multinationals partnering for research in developed countries painting the industries as problematic, an emphasis by Giulina as depicted by Schot and Steinmueller (2018) in linking with academia. Nevertheless, some universities are breaking these barriers and establishing partnerships to link with industry as this case study demonstrates. Both JKUAT and SU to different degrees and in different ways, have linked their research to research users. Both JKUAT and SU have not only tried to globalize their research and linking industry with academia, but have also tried to commodify these linkages through offering training programs that are needed for and by the society, e.g. Jomo Kenyatta University offers food technology courses (jam making, juicing) to County governments. These universities have also focused and built on their niches to advance linkage with the users

In particular, Strathmore has very strong culture and vision of working with the private sector and this enabled them to establish research centres specifically as collaboration nodes with industry. It has also developed non-traditional capability such as knowledge brokerage through other industrial players to enable this. Energy and climate innovation centres are particularly powerful in stimulating sustainability facing entrepreneurship, and other in driving economic success more generally. While performing better in these areas, there could

be greater incentives through public funding for the university to continue advancing the socio- economic and sustainability benefits it continues to leverage from its private sector collaboration.

JKUAT has done well in advancing partnerships in the agricultural sector. Although with less of a focus on formal industry partners, it has instead focused on communities and the SME sector and, because of the nature of agriculture, needs to follow a variety of innovation pathways with private sector led commercialisation but go beyond it. Collaboration with devolved county governments is particularly successful as training centres. However, the institution is yet to arrive at a model research – user interface to deploy the range of research products and expertise that it holds. Capacity building, institutional development and creating space for new models of partnership e.g. clear targeting of industry players in research agenda could help an organisation that is already well directed towards the sustainability agenda.

As indicated, governance of U-I linkages influences success and types of linkages pursued. The governance of U-I linkages as demonstrated in both the public and private universities is still not institutionalised through Knowledge transfer offices in terms of incentives for third mission actions although Geuna and Muscio (2009) argue that U–I collaborations have a more informal irrational management style than is often assumed. Nevertheless, a proper incentive mechanism need to exist beyond mentioning the individuals bringing partnerships for the case of JKUAT to proper structures that allow the researchers/lecturers dedicate time to compete for grants as well as other engagements.

In addition as demonstrated elsewhere, /focus on universities' performance contracts, and research assessment contracts may be effective tools to increase their accountability in establishing linkages.

What is emerging from these partnerships pursued are different knowledge systems innovation considered as new processes that could boost how university-industry linkages function and help re-direct interventions and investments to a better way of designing one. Strathmore's novelty lies in the fact that there are advances to make the University a business hub as well as an education hub because of the synergy this creates. The University is increasingly trying to embed industry within it in order to strengthen education industry synergy and research industry synergy. The university will benefit more from publicly funded innovation funds to direct success with business engagement and innovation towards social and sustainability issues. The JKUAT case points to a beginning of building U-I linkage that desires support and redesigning to commercialize the research innovations achieved.

6.0 REFERENCES

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